

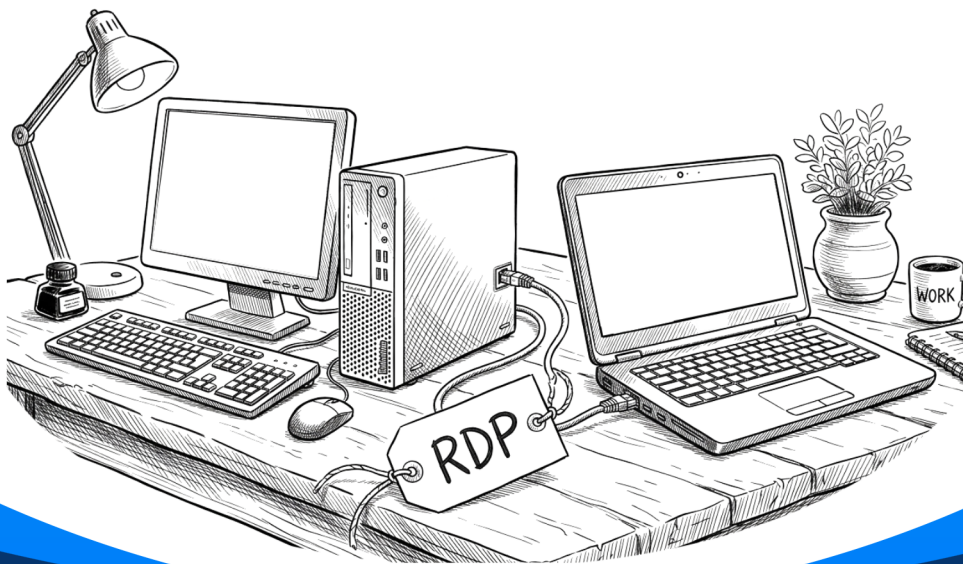
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Debian XFCE to Windows

Remote Desktop Connection

A Step -By-Step Guide to Set Up and Access Your Debian XFCE Desktop from Windows Using RDP



Install & Configure
xrdp on Debian

Set up XFCE
for Remote Access

Configure RDP
on Windows

Troubleshoot
Common Issues

Krythos

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1 Introduction

This guide explains how to access a **Debian 13 XFCE** desktop from a **Windows 11** PC using the built-in **Remote Desktop Connection** client (`mstsc.exe`).

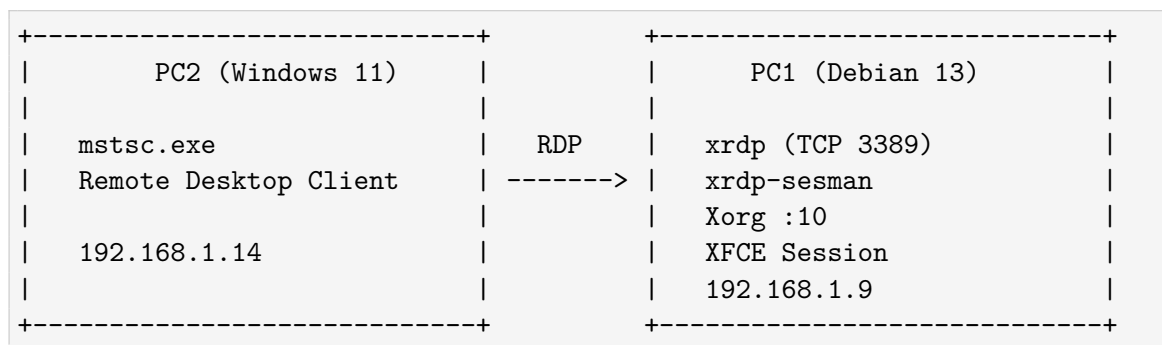
The setup uses **XRDP**, which is a free and open-source RDP server for Linux. It allows you to create a **separate graphical XFCE session** on the Debian machine that is independent of the physical desktop.

1.1 What You Will Achieve

After following this guide:

- The Debian PC (PC1) will run an XRDP server.
- The Windows PC (PC2) will connect to PC1 using Remote Desktop.
- You will see a **new, separate XFCE desktop** inside the Windows RDP window.
- The physical Debian desktop (if someone is sitting at it) will remain active and unaffected.

1.2 Architecture Overview



Note

The IP addresses 192.168.1.14 and 192.168.1.9 are examples. Replace them with the actual IP addresses of your own machines.

2 Prerequisites

Before starting, ensure the following:

1. **PC1 (Debian 13 XFCE)** is installed and working with a graphical desktop.
2. **PC2 (Windows 11)** is installed and working.
3. Both PCs are connected to the **same local network** (LAN/Wi-Fi).
4. You know the **username and password** for the Debian account you will use.
5. You have **sudo/root** access on the Debian PC.

6. You know the **IP address** of the Debian PC.

Tip

If you are unsure about the Debian IP address, you can find it later in Section 4. For now, just make sure both machines are on the same network.

3 Understanding XRDP

XRDP is an open-source implementation of the Microsoft RDP protocol. It allows a Linux machine to accept connections from any standard RDP client, including the one built into Windows.

3.1 How the Connection Works

```
Windows mstsc
|
| RDP (TCP 3389)
v
xrdp
|
v
xrdp-sesman
|
v
Xorg
|
v
XFCE Session
```

3.2 Key Components

- **xrdp** — The main service that listens for incoming RDP connections on port 3389.
- **xrdp-sesman** — The session manager that creates and manages the graphical session.
- **Xorg / xorgxrdp** — Provides the virtual X display for the RDP session.
- **startxfce4** — The command that starts the XFCE desktop.
- **dbus-run-session** — Ensures the RDP session gets its own DBus session, which is required for XFCE to start correctly.

4 Step 1 — Find the Debian PC's IP Address

On **PC1 (Debian)**, open a terminal and run:

```
hostname -I
```

Alternatively, you can use:

```
ip addr
```

Look for an address like:

```
192.168.1.9
```

Note

Write this IP address down. You will need it when connecting from Windows. In this guide, we will use 192.168.1.9 as the example Debian IP.

5 Step 2 — Update Debian

On **PC1 (Debian)**, open a terminal and update the package lists:

```
sudo apt update
```

Optionally, upgrade installed packages:

```
sudo apt upgrade -y
```

Tip

Keeping your system updated ensures you get the latest security patches and package versions.

6 Step 3 — Install XRDP and Required Packages

6.1 Install XRDP

```
sudo apt install xrdp -y
```

6.2 Install DBus Support

XRDP requires DBus to create a proper desktop session:

```
sudo apt install dbus-x11 -y
```

6.3 Verify XFCE Is Installed

Most Debian XFCE installations already have XFCE. Verify:

```
command -v startxfce4
```

Expected output:

```
/usr/bin/startxfce4
```

If the command is not found, install XFCE:

```
sudo apt install xfce4 xfce4-goodies -y
```

6.4 Verify dbus-run-session

```
command -v dbus-run-session
```

Expected output:

```
/usr/bin/dbus-run-session
```

If it is missing, reinstall dbus-x11:

```
sudo apt install --reinstall dbus-x11 -y
```

7 Step 4 — Enable and Start the XRDP Service

Enable XRDP to start automatically at boot and start it immediately:

```
sudo systemctl enable --now xrdp
```

7.1 Check the Service Status

```
systemctl status xrdp
```

You should see:

```
Active: active (running)
```

Also check the session manager:

```
systemctl status xrdp-sesman
```

Tip

Press **q** to exit the status view if it opens a pager.

8 Step 5 — Verify XRDP Is Listening on Port 3389

Run:

```
sudo ss -lntp | grep 3389
```

You should see output similar to:

```
LISTEN 0 2 0.0.0.0:3389 0.0.0.0:* users:(("xrdp",pid=1234,fd=11))
```

Note

TCP port 3389 is the standard RDP port. If you do not see a listening socket, check the service status and logs:

```
systemctl status xrdp
sudo journalctl -u xrdp --no-pager
```

9 Step 6 — Configure the XRDP Startup Script

This is the most important configuration step. The file `/etc/xrdp/startwm.sh` controls what happens after a user successfully authenticates.

9.1 Back Up the Original File

```
sudo cp /etc/xrdp/startwm.sh /etc/xrdp/startwm.sh.backup
```

9.2 Edit the File

```
sudo nano /etc/xrdp/startwm.sh
```

9.3 Replace the Contents

Delete everything in the file and replace it with the following:

```
#!/bin/sh

if [ -r /etc/profile ]; then
    . /etc/profile
fi
```

```
if [ -r "$HOME/.profile" ]; then
    . "$HOME/.profile"
fi

unset DBUS_SESSION_BUS_ADDRESS
unset XDG_CURRENT_DESKTOP
unset XDG_SESSION_DESKTOP
unset DESKTOP_SESSION

exec dbus-run-session -- startxfce4
```

9.4 Save and Exit

In nano, press:

```
Ctrl + O
Enter
Ctrl + X
```

9.5 Make the Script Executable

```
sudo chmod +x /etc/xrdp/startwm.sh
```

Verify the permissions:

```
ls -l /etc/xrdp/startwm.sh
```

You should see:

```
-rwxr-xr-x 1 root root ... /etc/xrdp/startwm.sh
```

Note

Why this configuration?

- `unset DBUS_SESSION_BUS_ADDRESS` — Prevents the RDP session from inheriting the physical desktop's DBus session.
- `unset XDG_CURRENT_DESKTOP`, etc. — Clears desktop environment variables that could conflict.
- `exec dbus-run-session - startxfce4` — Creates a fresh DBus session and starts XFCE inside it. This ensures a clean, separate desktop.

10 Step 7 — Remove Any Conflicting User Session File

If a file called `~/.xsession` exists, it can interfere with the XRDP startup process. Remove it:

```
rm -f ~/.xsession
```

Verify it is gone:

```
ls -la ~/.xsession
```

You should see:

```
ls: cannot access '/home/user01/.xsession': No such file or directory
```

11 Step 8 — Restart XRDP

After making configuration changes, restart the XRDP service:

```
sudo systemctl restart xrdp
```

Check that it is running:

```
systemctl status xrdp
```

Also check the session manager:

```
systemctl status xrdp-sesman
```

12 Step 9 — Check the Debian Firewall (If Enabled)

If you have a firewall enabled (such as `ufw`), you need to allow TCP port 3389.

12.1 Check if ufw Is Active

```
sudo ufw status
```

12.2 Allow RDP Port

```
sudo ufw allow 3389/tcp
```

12.3 Reload the Firewall

```
sudo ufw reload
```

Note

If `ufw` is not installed or not active, you can skip this step. Debian does not enable a firewall by default.

13 Step 10 — Test the Connection from Windows

Now switch to **PC2** (Windows 11).

13.1 Open Remote Desktop Connection

Press:

```
Win + R
```

Type:

```
mstsc
```

Press **Enter**.

13.2 Enter the Debian IP Address

In the **Computer** field, enter the Debian PC's IP address:

```
192.168.1.9
```

Click **Connect**.

13.3 Enter Credentials

When prompted, enter:

```
Username: user01  
Password: <your Debian password>
```

Note

Replace `user01` with your actual Debian username.

13.4 Choose the Session Type

XRDP may ask you to choose a session type. Select:

```
Xorg
```

13.5 Expected Result

You should now see a **new XFCE desktop** inside the Windows Remote Desktop window.

Tip

If the connection fails, proceed to Section 12 (Troubleshooting) before continuing.

14 Step 11 — Verify the Setup

14.1 Check the Display Number

Inside the **physical Debian desktop**, open a terminal and run:

```
echo $DISPLAY
```

Expected output:

```
:0
```

Inside the **Windows RDP session**, open a terminal and run:

```
echo $DISPLAY
```

Expected output:

```
:10
```

Note

The exact XRDP display number may be :10, :11, etc. The important thing is that it is **different** from the physical :0 display.

14.2 Check Logged-In Sessions

On Debian, run:

```
loginctl
```

You can also use:

```
who  
w
```

14.3 Check XFCE Processes

```
pgrep -a xfce
```

This should show XFCE processes for both the physical session and the RDP session.

15 Troubleshooting

15.1 Problem: Connection Refused

Symptoms: Windows says it cannot connect to the remote computer.

Checks:

```
systemctl status xrdp  
sudo ss -lntp | grep 3389
```

Fix:

```
sudo systemctl restart xrdp
```

Then try again.

15.2 Problem: Windows Cannot Reach Port 3389

On Windows, open PowerShell and run:

```
Test-NetConnection 192.168.1.9 -Port 3389
```

Look for:

```
TcpTestSucceeded : True
```

If it is **False**, check:

- Debian firewall (see Section 9)
- Both PCs are on the same network
- The Debian IP address is correct
- XRDP is running

15.3 Problem: Authentication Succeeds but Desktop Closes Immediately

Check the logs:

```
sudo journalctl -u xrdp-sesman --since "5 minutes ago" --no-pager
sudo journalctl -u xrdp --since "5 minutes ago" --no-pager
```

Verify:

```
command -v startxfce4
command -v dbus-run-session
cat /etc/xrdp/startwm.sh
```

Make sure the final line of `startwm.sh` is:

```
exec dbus-run-session -- startxfce4
```

Also remove any conflicting user session file:

```
rm -f ~/.xsession
```

Then restart:

```
sudo systemctl restart xrdp
```

15.4 Problem: Exit Code 127

Exit code 127 means “command not found”.

Check:

```
command -v startxfce4
cat /etc/xrdp/startwm.sh
ls -la ~/.xsession
```

Look for typos such as `startxfce4startxfce4` instead of `startxfce4`.

15.5 Problem: Window Manager Exited Quickly

Check the logs:

```
sudo journalctl -u xrdp-sesman --since "5 minutes ago" --no-pager
```

Check XFCE processes:

```
pgrep -a xfce
```

Verify the startup script:

```
cat /etc/xrdp/startwm.sh
```

Remove conflicting session file:

```
rm -f ~/.xsession
```

Restart:

```
sudo systemctl restart xrdp
```

15.6 Problem: Black Screen After Login

- Wait 10–15 seconds. XFCE may take a moment to start.
- Check the logs (see above).
- Ensure `dbus-run-session` is installed.
- Ensure the `startwm.sh` file is correct.
- Try disconnecting and reconnecting.

16 Quick Reference

16.1 Debian Commands

```
# Find IP address
hostname -I

# Check XRDP status
systemctl status xrdp

# Restart XRDP
sudo systemctl restart xrdp

# Enable XRDP at boot
sudo systemctl enable xrdp

# Check listening port
sudo ss -lntp | grep 3389

# Verify XFCE
command -v startxfce4

# Verify DBus
command -v dbus-run-session

# Show current display
```

```
echo $DISPLAY

# List sessions
loginctl

# Show XFCE processes
pgrep -a xfce

# View XRDP logs
sudo journalctl -u xrdp --since "5 minutes ago" --no-pager

# View session manager logs
sudo journalctl -u xrdp-sesman --since "5 minutes ago" --no-pager
```

16.2 Windows Commands

```
Win + R
mstsc
192.168.1.9
Xorg
user01
<Debian password>
```

16.3 Key Configuration File

```
/etc/xrdp/startwm.sh
```

Final contents:

```
#!/bin/sh

if [ -r /etc/profile ]; then
    . /etc/profile
fi

if [ -r "$HOME/.profile" ]; then
    . "$HOME/.profile"
fi

unset DBUS_SESSION_BUS_ADDRESS
unset XDG_CURRENT_DESKTOP
unset XDG_SESSION_DESKTOP
unset DESKTOP_SESSION

exec dbus-run-session -- startxfce4
```

17 Security Considerations

Warning

Do NOT expose RDP directly to the Internet.

XRDP should only be used on a trusted local network. Never forward TCP port 3389 from your router to the Debian PC unless you fully understand the security risks.

Safe setup:

```

Windows PC
  |
  | Local LAN only
  v
Debian XRDP

```

Unsafe setup:

```

Internet
  |
  v
Router :3389 forwarded
  |
  v
Debian XRDP

```

If you need remote access from outside your home network, use a **VPN** instead.

17.1 Additional Security Tips

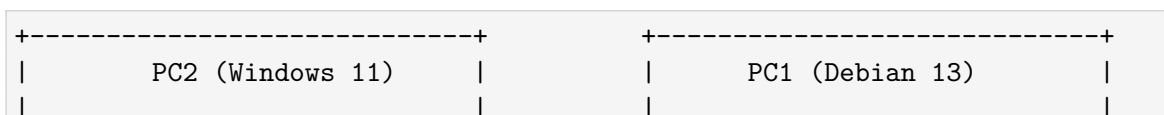
- Use strong passwords for the Debian account.
- Keep Debian and XRDP updated.
- Consider restricting XRDP access to specific IP addresses using firewall rules.
- Disable XRDP when not needed:

```
sudo systemctl stop xrdp
```

18 Summary

You have successfully set up XRDP on Debian 13 XFCE and connected to it from Windows 11 using the built-in Remote Desktop client.

18.1 What Was Achieved



	mstsc.exe		RDP		xrdp (TCP 3389)	
	Remote Desktop Client		----->		xrdp-sesman	
					Xorg :10	
	192.168.1.14				XFCE Session	
					192.168.1.9	
+	-----+				-----+	+

18.2 Key Points

- The physical Debian desktop remains at `DISPLAY=:0`.
- The RDP session runs at `DISPLAY=:10` (or similar).
- Both sessions run simultaneously under the same user.
- The critical configuration file is `/etc/xrdp/startwm.sh`.
- The key startup line is:

```
exec dbus-run-session -- startxfce4
```

18.3 Next Steps

- Test the connection from other devices on your LAN.
- Adjust XFCE settings inside the RDP session as needed.
- Consider setting up a VPN for remote access.
- Keep your system updated for security.

End of Guide
